

GOVERNMENT COLLEGE UNIVERSITY LAHORE

Department of Electronics

INDUSTRIAL ELECTRONICS LAB REPORT

Sensor Based Navigation Module

Arduino Based Obstacle Avoiding Car

Course	Industrial Electronics Lab
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Project	Arduino Based Obstacle Avoiding Car

1 Title

Arduino Based Obstacle Avoiding Car

The Arduino Based Obstacle Avoiding Car is an autonomous robotic vehicle that can detect obstacles in its path and automatically change its direction. The system uses an HC-SR04 ultrasonic sensor to measure the distance of an object. Arduino Uno R3 processes the sensor signal and controls two DC geared motors through an L298N motor driver.

2 Components Used

No.	Component	Purpose
1	Arduino Uno R3	Main controller of the robot. It reads the sensor and gives control signals to the motor driver.
2	HC-SR04 Ultrasonic Sensor	Detects obstacles and measures their distance.
3	L298N Motor Driver	Controls the direction of the two DC motors.
4	2 × DC TT Geared Motors	Provide movement to the left and right wheels.
5	Robot Chassis and Wheels	Provide the mechanical structure and movement of the car.
6	Battery / Power Supply	Provides power to the Arduino and motor circuit.
7	Jumper Wires	Used to make electrical connections.

The main components required for this project are Arduino Uno R3, HC-SR04 ultrasonic sensor, L298N motor driver and two DC TT geared motors. These components work together to provide sensing, processing and movement.

3 Theory

3.1 Arduino Uno R3

Arduino Uno R3 is a microcontroller development board based on the ATmega328P. It is used to read input signals from sensors and control output devices. In this project, Arduino receives the distance information from the HC-SR04 sensor and sends logic signals to the L298N motor driver.

3.2 HC-SR04 Ultrasonic Sensor

The HC-SR04 is an ultrasonic distance measuring sensor. It has four pins: VCC, TRIG, ECHO and GND. Arduino sends a short trigger pulse to the TRIG pin. The sensor sends ultrasonic waves toward an object. When the waves are reflected back, the ECHO pin produces a pulse. Arduino measures the duration of this pulse and calculates the distance.

The distance is calculated using the following relation:

$$\text{Distance} = \frac{\text{Time} \times \text{Speed of Sound}}{2}$$

In the program, the approximate speed of sound is taken as $0.034 \text{ cm}/\mu\text{s}$. The division by 2 is used because the ultrasonic wave travels from the sensor to the object and then returns to the sensor.

3.3 L298N Motor Driver

The L298N is a dual H-bridge motor driver. It is used because Arduino pins cannot directly provide the high current required by DC motors. The L298N receives logic signals from Arduino and controls the direction of the two motors. IN1 and IN2 control the left motor, while IN3 and IN4 control the right motor.

3.4 DC Geared Motors

DC geared motors convert electrical energy into mechanical rotation. The gearbox reduces the motor speed and increases torque. This makes the motors suitable for a small robot car. One motor is connected to the left side and the other motor is connected to the right side.

3.5 Obstacle Avoidance

The robot continuously measures the distance in front of it. In this project, 20 cm is selected as the obstacle distance. When the path is clear, both motors move the car forward. When an object is detected within 20 cm, the car stops, moves backward for a short time, and then turns right to avoid the obstacle.

4 Code

```
1 // Define Ultrasonic Sensor Pins
2 #define TRIG_PIN 10
3 #define ECHO_PIN 11
4 // Define L298N Motor Driver Logic Pins
5 #define IN1 4
6 #define IN2 5
7 #define IN3 6
8 #define IN4 7
9 // Define Obstacle distance threshold in cm
10 #define OBSTACLE_DISTANCE 20
11
12 void setup() {
13     Serial.begin(9600);
14     pinMode(TRIG_PIN, OUTPUT);
15     pinMode(ECHO_PIN, INPUT);
16     pinMode(IN1, OUTPUT);
17     pinMode(IN2, OUTPUT);
18     pinMode(IN3, OUTPUT);
19     pinMode(IN4, OUTPUT);
20     delay(2000);
21 }
22
23 void loop() {
24     int distance = getDistance();
25     Serial.print("Distance: ");
26     Serial.print(distance);
27     Serial.println(" cm");
28
29     if (distance > 0 && distance < OBSTACLE_DISTANCE) {
30         stopMotors();
31         delay(300);
32         moveBackward();
33         delay(500);
34         stopMotors();
35         delay(300);
36         turnRight();
37         delay(400);
38     } else {
39         moveForward();
40     }
41     delay(50);
42 }
43
44 int getDistance() {
45     digitalWrite(TRIG_PIN, LOW);
46     delayMicroseconds(2);
47     digitalWrite(TRIG_PIN, HIGH);
48     delayMicroseconds(10);
49     digitalWrite(TRIG_PIN, LOW);
50
51     long duration = pulseIn(ECHO_PIN, HIGH, 30000);
52     if (duration == 0) {
53         return 100;
54     }
55     return duration * 0.034 / 2;
```

```

56 }
57
58 void moveForward() {
59     digitalWrite(IN1, HIGH);
60     digitalWrite(IN2, LOW);
61     digitalWrite(IN3, HIGH);
62     digitalWrite(IN4, LOW);
63 }
64
65 void moveBackward() {
66     digitalWrite(IN1, LOW);
67     digitalWrite(IN2, HIGH);
68     digitalWrite(IN3, LOW);
69     digitalWrite(IN4, HIGH);
70 }
71
72 void turnRight() {
73     digitalWrite(IN1, HIGH);
74     digitalWrite(IN2, LOW);
75     digitalWrite(IN3, LOW);
76     digitalWrite(IN4, HIGH);
77 }
78
79 void stopMotors() {
80     digitalWrite(IN1, LOW);
81     digitalWrite(IN2, LOW);
82     digitalWrite(IN3, LOW);
83     digitalWrite(IN4, LOW);
84 }

```

The program first initializes the ultrasonic sensor and motor driver pins. It then continuously measures the obstacle distance. If the distance is 20 cm or more, the car moves forward. If the distance is less than 20 cm, the car stops, moves backward, stops again and turns right.

5 Procedure

1. Assemble the robot chassis and attach the two DC TT geared motors and wheels.
2. Mount the HC-SR04 ultrasonic sensor at the front of the car.
3. Connect HC-SR04 VCC to 5V and GND to GND. Connect TRIG to Arduino D10 and ECHO to Arduino D11.
4. Connect L298N IN1, IN2, IN3 and IN4 to Arduino D4, D5, D6 and D7 respectively.
5. Connect the left motor to the L298N motor output terminals and connect the right motor to the other output terminals.
6. Connect the suitable battery or power supply to the motor driver and make sure the circuit has a common ground where required.
7. Open Arduino IDE and enter the given program.
8. Select Arduino Uno board and the correct COM port, then upload the program.

9. Place the robot on a flat surface and switch on the power supply.
10. The robot waits for about two seconds and then starts moving forward.
11. Place an object in front of the robot. When the distance becomes less than 20 cm, observe the automatic obstacle avoidance.
12. Check the distance values through the Serial Monitor at 9600 baud.

6 Conclusion

The Arduino Based Obstacle Avoiding Car was designed using an Arduino Uno R3, HC-SR04 ultrasonic sensor, L298N motor driver and two DC geared motors. The ultrasonic sensor detects an obstacle and sends distance information to the Arduino. Arduino processes this information and controls the motors through the L298N motor driver.

When the path is clear, the car moves forward. When an obstacle is detected within 20 cm, the car stops, moves backward and turns right automatically. This experiment demonstrates the basic concept of sensor-based navigation and autonomous robot control.

The project can be further improved by adding a servo motor for scanning different directions, IR sensors for additional detection, PWM-based speed control, and more advanced path-planning techniques.